

# Requirements Study for a Professional Green Coding Course Format

## A Focus Group Study to Explore Strategies for Implementing Green Coding Training in Software Development

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### Abstract:

Although Green Coding (GC) is increasingly discussed in academia and, to some extent, within the industrial sector, it has not yet become a standard part of software development across the IT industry. GC focuses on reducing the environmental impact of software throughout its lifecycle, from production to use and disposal. To achieve this, it is essential to integrate GC knowledge and practices into the software development process. Previous work has explored integration into academic curricula and how junior developers can introduce this knowledge into the industry. In contrast, this study focuses on bringing GC knowledge directly into the industrial environment, addressing key challenges and discussing suitable training formats.

The methodology applied was a semi-structured expert interview conducted in the form of a focus group during a workshop at EnviroInfo 2024 at the GUC in Cairo.

The expert workshop revealed that the successful adoption of Green Coding depends on its integration into existing software development processes, tools, and training structures. Among various strategies discussed, training emerged as a central success factor. In-house training was considered the most effective format, as it enabled interactive learning, peer exchange, and a stronger commitment to behavioral change. Professional gatherings and workshops also play a crucial role in fostering expert exchange and continuous knowledge transfer. While online courses and webinars offer scalability, they were perceived as less impactful unless combined with certification or interactive components. Certificates, particularly those recognized by industry, were viewed as valuable incentives that could increase the perceived relevance of GC skills in the job market. However, their success depends on the recognition and integration of the broader market into recruitment practices. The integration of GC topics into academic curricula and technical training programs, either as standalone courses or embedded content, was highlighted as an essential long-term measure. Participants emphasized the need for regular updates to the content of GC training, ideally driven by expert communities, conferences, and changes in development frameworks. Overall, the workshop participants advocated for a flexible, multi-format training strategy. Combining in-house workshops, formal certification, academic courses, and integration into daily tools and processes ensures that GC knowledge reaches both current professionals and future developers. Training must be practical, context-sensitive, and regularly updated to remain effective and support a sustainable transformation in the software industry.

**Keywords:** Green Coding, Software Engineering, Requirements, Workshop Review, Focus Group, Green IT, Green Computing, Sustainable Software Engineering, Course, Training

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## 1 Introduction and Motivation

The 'Potentials of Green Coding' [In24] research project explored environmentally sustainable programming practices and their integration into software development processes and educational curricula. This study included a comprehensive literature review, analysis of industry practices, and validation through workshops and expert interviews. In the following project, 'Unlocking Potentials of Green Coding into Practice', the recommendation to put these mechanics into practice was used to develop them. The third research part of the project addresses the lack of GC knowledge among practitioners, identified in previous studies, as [He23] or also in part of the Literature Report as [Ju23]. However, the research identified practical challenges, emphasizing the need to train not only junior developers but also those already working in the industry. The previous research was focused on integrating the knowledge of GC in possible university curricula as in [Ju25a], due to the most (71, 5%) of software development professionals worldwide who had a university degree (41% bachelor's, 25.6% master's and 4.9% professional degrees like Ph.D., while 12.6% had some university education without earning a degree. In contrast, 11.6% of developers entered the field without a university background [St24]. This resulted in the creation of 'Academic Guidelines and Recommendations for Integrating Green Coding Aspects into Curricula of Higher Education' [Ju25a], which outlines materials and methods for embedding GC into the education of future software developers. The first step was to find out what knowledge is most lacking for young professionals with a GC background (either educational or self-taught [JW24] to A) develop a training concept that can be applied broadly across Information and Communications Technology (ICT) and software development fields and B) to bring this together with the methodologies and content of GC in the professional field through training.

This paper is a step in the process where we aim to evaluate possible materials and a training concept to address this knowledge gap. We strive to evaluate and identify the most effective training formats. To find and create a training form, training methodologies, and content, and in the third step by developing a training course that is implemented and evaluated on a test basis. The aim is to make this training available to the public.

The article is structured first to present the motivation and the current state of research. This is followed by an explanation of the methodology and tools used for the focus group. The evaluation section discusses the findings. Based on these results, clear recommendations are provided in the conclusion section for integrating GC into training programs.

## 2 Methodology

To investigate the opportunities and challenges of integrating GC methods into software engineering, a focus group [Ki95] approach was selected as the central qualitative methodology. This choice was based on the need for in-depth, interactive discussions that could reveal nuanced perspectives on training formats, content relevance, and practical implementation

issues across different sectors and roles. The authors followed the approach ideas of [Ro20] and [MM].

The Chair-Guided Structured Expert Interviews, particularly the Focus Groups method, were chosen for several reasons:

- It allows direct exchange between participants from diverse backgrounds, enabling the exploration of varied expectations and experiences.
- It is well-suited to early-stage idea development, especially for a complex and evolving topic like GC.
- It supports a collaborative setting, in which group dynamics can surface shared challenges and innovative solutions that individual interviews might miss.

The primary aim was to collect input for the development of an internationally applicable GC training course. Therefore, the group was formed during an international conference, ensuring geographic and professional diversity. There were 3 online and 10 in-house participants, including:

- Doctoral students
- Academics with links to industry
- Full-time researchers
- Industry practitioners
- Policymakers with ties to sustainable digitalization

Selection criteria focused on assembling a mix of individuals already engaged or interested in sustainable software development, ensuring both depth and relevance in the discussion.

## **2.1 Acquisition Methodology**

The Focus Group was conducted on Wednesday, 13.11.2024, on the EnviroInfo Conference [Ge24] in Cairo as the last session of the 'Green Coding Techniques and Methodologies' track. The track contained 6 paper presentations before, and the workshop was announced throughout the day. The workshop was titled: 'Workshop Unlocking the Potentials of GC into Practice - Focus Groups on: A) Labeling Schemes as Tools for Software Sustainability B) Opportunities and Challenges of Integrating GC Methods in Software Engineering Processes' and contained the first three questions for Topic A, corresponding to the research package of the Umweltcampus Birkenfeld, presented and moderated by Mr. Westing followed by the three questions presented by the author for the working package of the HTW Berlin. The focus groups were time-boxed to 5 minutes each question.

## 2.2 Evaluation Methodology

The focus group was overseen by two individuals, one of whom was responsible for facilitating the session in person, while the other oversaw the online segment of the event. To ensure clarity and inclusivity for all participants, microphones were the sole form of communication, distributed to the speakers for their use. The video and audio excerpts for this presentation were recorded by the event organizer. The present recording has undergone conversion from the video format MP4 to the audio format MP3. The current iteration of the MP3 has undergone a reduction in length, with the result that only questions 4-6 of [Ju25c] are addressed. These questions (3.1 to 3.3) are the ones pertinent to the present document. The transcription of the file was subsequently executed using the Turboscribe [Tu25] page, with the speakers' details and time stamps recorded for each speaker. Subsequently, the data was imported into Google NotebookLM [Go25] as the only source so that the conversation tendency and the main topics could be analyzed and identified. In addition, a text file was added with the information on the PowerPoint slides used during the focus group as a reference.

## 3 Evaluation and Discussion

The complete anonymized transcript of the audio and the transcriber's notes can be found in the 'supplementary material A' at <https://doi.org/10.5281/zenodo.15729451> [Ju25c]. We also provide a mindmap addressing all the following topics under <https://doi.org/10.5281/zenodo.15728863> [Ju25b]. Even though the evaluation of the focus groups deals exclusively with the opinions and ideas of the participants, additional references have been inserted by the author at unclear points in order to provide the reader with a clearer understanding of the context discussed.

### 3.1 Challenges and Strategies in Integrating Green Coding Practices

The integration of GC practices into existing software development processes presents several significant challenges for companies, alongside potential strategies to overcome these obstacles. This discussion synthesizes expert perspectives on these issues. The Question asked to start this time-box was: 'What are the main challenges that companies face when integrating Green Coding practices into existing software development processes, and what strategies can be used to overcome these obstacles?'

#### 3.1.1 Main Challenges Identified:

A primary challenge is the lack of awareness among software developers regarding the energy consumption implications of their code and underlying infrastructure. Specifically, there is

a lack of knowledge about inefficiencies in data centers, such as the significant idling of servers. Furthermore, companies often prioritize value creation and rapid revenue generation over sustainability initiatives. Sustainability may not be embedded within the core corporate strategy, remaining merely a recommendation rather than a directive. Another obstacle is the difficulty for companies to train their sophisticated models (e.g., for code classification or assessment) due to limited capacity. There is a clear need for accessible pre-trained models. While knowledge about GC exists, it is not easily accessible or consolidated in one central location. This fragmentation makes it hard for developers to find and apply relevant information. Finally, sustainability is often not yet treated as a formal functional or non-functional requirement within the software development process.

### 3.1.2 Strategies for Overcoming These Obstacles:

Several strategies were proposed to address these challenges:

**Enhance Awareness:** Actively work to create awareness among developers about the importance and impact of GC. Providing catchy examples that demonstrate the significant scaling effect of software on energy consumption can highlight why developers should invest effort in this area.

**Leverage Gamification:** Introduce gamification elements, such as challenges, high score boards, or transparent 'blackboards', to motivate developers and foster healthy competition in optimizing code for energy efficiency. Even when the part of the focus group in this context was quite brief, the author refers to the positive outcomes of gamification in other learning environments, as described in [BRW19].

**Develop and Integrate Tooling:** Tooling is considered a crucial strategy, particularly focusing on measurement. Tools can provide developers with real-time feedback on code 'greenness' directly within their Integrated Development Environment, offering recommendations for improvement. Tools can also assess the energy consumption of software architectures and suggest optimizations. Solutions such as co-pilot [Mi25] were mentioned as potentially contributing to energy reduction. Providing visualization of Key Performance Indicators related to energy consumption can make progress transparent for both developers and management.

**Provide Accessible Pre-trained Models:** Publishing pre-trained models related to GC (e.g., for code analysis or assessment) on accessible platforms like Machine Learning as a Service or AI as a Service hubs (similar to Hugging Face [Hu25]) would allow companies to reuse them without needing to train their own. Such hubs could include KPIs on the model's effectiveness or environmental sustainability to aid in selection. This promotes code reusability and scaling effects.

**Establish a Reputable Body of Knowledge:** Consolidate existing knowledge on GC practices into a reputable 'Body of Knowledge'. This knowledge should be easily accessible, possibly through comprehensive guidelines similar to web standards for green web development. Concepts like 'Energy-Safe by Design' or 'Low Energy Consumption by Design' and sustainable design patterns that offer solutions to recurring energy consumption problems could be developed and disseminated.

**Integrate Sustainability into Company Strategy:** Elevate sustainability from a recommendation to a mandatory component of corporate strategy.

**Define Sustainability as a (Non-)Functional Requirement:** Formally incorporate sustainability as a (non-)functional requirement within the software development process (as suggested for example, by [Ra14]). This includes integrating it into the requirements engineering phase. Utilizing artifacts from models like a Green Software Engineering Model (e.g., the GREENSOFT Model [NKD14]), such as sustainability journals or introspection techniques, can support this integration within methodologies like Scrum or traditional processes.

**Integrate Green Coding into Existing Frameworks:** Embed GC patterns, philosophy, and practices directly into widely used software development frameworks. Updating these frameworks would force developers to learn and apply GC principles, ensuring the content remains current.

### 3.1.3 Results of the Expert Voting:

The participants voted on what topic would be most important to focus on in the integration process. The options were soft skills (change management), measurement (tools), and software engineering process information. In the vote, five participants were in favor of Measurement (Tools) and seven in favor of Software Engineering Process Information, while Soft Skills (Change Management) received one vote.

This indicates that the experts consider the areas of tooling (measurement) and integration into the software development process (software engineering process information) to be particularly important for overcoming the challenges and implementing strategies in GC.

## 3.2 Possible Training Formats

Training programs for GC can be tailored to address the specific needs and challenges faced by different roles in software engineering, including developers, project managers, and quality assurance engineers. The primary objective of these programs is to support the practical integration of GC practices within companies. Starting with the focus group question, 'How can training programs be designed to address the specific needs and

challenges faced by different roles in software engineering, such as developers, project managers, and quality assurance engineers? What format is preferred?', multiple formats and approaches were proposed to ensure effective delivery of the training:

### 3.2.1 Suitable Training Formats as Described in the Focus Group

**Face-to-face and In-house Training:** This approach is considered highly effective for fostering a shared understanding and cultivating a cultural shift towards GC within an organization. Participants are more likely to internalize the importance of the topic when engaging in face-to-face interactions, workshops, and discussions, which can also help address anxieties and encourage a collective commitment. In-house training specifically provides an environment where individuals can learn alongside like-minded colleagues, enhancing the learning experience. It is also crucial for sensitizing employees and convincing leadership about the benefits of GC.

**Certificates and Online Tests with Certifications:** The provision of certificates is seen as a strong motivator for individuals to actively engage with and study the content of the GC, rather than simply passively consuming information. Certificates can also enhance a professional's curriculum vitae and potentially encourage recruiters and companies to prioritize candidates possessing certified GC skills, although motivating recruiters to focus on such certifications remains a challenge. A formal certification program, such as becoming a 'certified green coder', could serve as a foundational step.

**Conferences and Gatherings:** These events offer a valuable platform for disseminating knowledge, sharing methodologies, and presenting up-to-date research findings related to GC. They facilitate networking and information exchange among specialists from different companies and research institutions, promoting a wider adoption of GC practices.

**Integration into Academic Curricula:** From an academic perspective, it is recommended that GC topics be systematically integrated into existing university curricula for software engineering and DevOps. Offering GC courses as electives can provide students with optional pathways to acquire these essential skills.

**Webinars and Online Courses:** While mentioned as a potential format, some participants expressed concerns that online-only methods, such as clicking through slides, might not effectively convey the deeper significance or foster a tangible 'feeling' for the topic.

### 3.2.2 Results of the Voting Regarding Training Formats

During the discussion on preferred training formats, a vote was conducted offering the following options: Webinar / Online Course, In-house Training, and Online Test with Certificate. The results indicated a clear preference for in-house training, which received

the highest number of votes (7). This was followed by spontaneously added conferences and gatherings (6 votes), certificate-based online tests (3 votes), and webinars or online courses (2 votes).

These results reflect a general inclination towards interactive and presence-based learning formats.

### 3.3 Integrating and Updating Green Coding in Technical Training

This focus group section aimed to identify strategies for meaningfully integrating GC training into existing technical training programs and ensuring its long-term relevance. The following sections summarize the discussed approaches.

Key guiding questions included: 'How can Green Coding training be integrated into existing technical training programs without overwhelming developers with additional requirements?' and 'How can we ensure that GC content remains up to date? What are suitable forms of integration?'

#### 3.3.1 Ideas of the Focus Group of possible Integrations

**Integration into Existing Technical Training Programs:** Training programs for GC can be integrated into existing technical training programs and maintained for up-to-date content through several proposed strategies, to prevent developers from being overwhelmed by additional requirements.

**Leveraging Development Frameworks for Green Coding Principles:** A particularly effective strategy involves embedding GC patterns, foundational philosophies, and related principles directly into existing development frameworks that are widely utilized by software professionals. This method inherently encourages developers to adopt coding and thinking practices aligned with GC, as updates to the framework would necessitate continuous learning and adherence to current GC standards.

**Motivating Developers through Practical Examples and Impact Awareness:** It is crucial to present compelling and impactful examples to developers, illustrating the practical benefits of investing in GC, especially considering the substantial scaling effect of software. By fostering sufficient awareness of these advantages, developers may be intrinsically motivated to proactively seek out and engage with relevant GC material independently.

**Ensuring Green Coding Content Stays Up-to-Date:** To ensure the continuous relevance and currency of GC content, conferences and specialized gatherings are identified as vital platforms for disseminating the latest research findings and methodological advancements. These events could feature dedicated 'GC for specialists' days or tracks, fostering valuable exchange and collaboration among experts from various companies and research institutions.



**Role of Conferences and Expert Gatherings:** Conferences and professional gatherings offer essential opportunities to share current findings, methodologies, and best practices. The inclusion of specialized sessions focused on GC ensures ongoing engagement with the latest developments.

**Framework Updates as a Mechanism for Continuous Learning:** The integration of GC principles into actively maintained and updated development frameworks inherently contributes to the currency of the content, as developers would continuously learn about new GC developments as part of routine framework updates.

### 3.3.2 Fitting Integration Forms (Vote Results)

In the discussion on suitable integration forms for GC training, a vote was conducted with the following options: Stand-alone Training, Full Integration into Courses, Other Options, and Both/All Options. The most favored approach was Both/All Options, receiving six votes. Full Integration into Courses followed with four votes, while Stand-alone Trainings received three votes. Other Options did not receive any votes.

These results suggest a preference for flexible and combined training approaches.

## 4 Conclusion

The findings of this expert workshop highlight a clear need for the structured integration of GC practices into the broader software engineering domain. Experts emphasized that successful implementation depends not only on raising awareness but also on embedding GC principles into existing software development processes, tools, and training programs. The voting results underline the importance of practical integration—particularly through updates in development frameworks and the use of measurement tools—as well as the relevance of software engineering process information. Tooling and process alignment are seen as enablers that make sustainable development tangible and manageable in daily practice.

Training formats emerged as a key success factor. Interactive and presence-based formats, especially in-house training and professional gatherings, were strongly preferred over passive online formats. These approaches are more effective in creating shared understanding, encouraging behavior change, and fostering a sense of ownership. Certification programs were also mentioned as valuable motivators, though their effectiveness depends on market acceptance. Academic integration through elective courses and embedding GC content into existing technical curricula were seen as strategic entry points for long-term cultural change in the industry.

To remain relevant, GC content requires regular updates. This can be supported by expert communities, conferences, and knowledge sharing. Integrating GC principles into widely used software development frameworks helps ensure that developers encounter sustainable practices as part of their routine work, making sustainability a standard consideration in software engineering.

Ultimately, the discussions and results indicate that there is no one-size-fits-all approach. A flexible combination of training formats and integration strategies—tailored to different organizational and individual needs—was the most supported direction. GC must become a cross-cutting concern that is interwoven with existing roles, responsibilities, and workflows in software development. Only then can its potential be fully realized to reduce the environmental impact of digital technologies in a scalable and lasting manner.

## Acknowledgments

This work was funded by the Internet Society Foundation<sup>2</sup>. The authors extend their sincere thanks to the EnviroInfo Conference at the German University in Cairo for providing the opportunity to conduct this expert interview as part of the workshop.

## References

- [BRW19] Bernik, A.; Radošević, D.; Wattanasoontorn, V.: Influence, principles and good practice of computer game elements: Mechanics and Dynamics. In: 2019 23rd International Computer Science and Engineering Conference (ICSEC). Pp. 241–245, 2019.
- [Ge24] German University in Cairo: EnviroInfo 2025- Transboundary Environmental Challenges: Digital Inclusion for Sustainable Development, 2024, <https://enviroinfo.org/>, accessed: 06/07/2025.
- [Go25] Google: NotebookLM, 2025, <https://notebooklm.google.com/>, accessed: 06/05/2025.
- [He23] Henze, C. et al.: Potentials of Green Coding: Perspectives on Implementation, Berlin, 2023, <https://gi.de/meldung/green-coding-publikation>, accessed: 06/05/2025.
- [Hu25] Hugging Face, I.: Hugging Face Hub, <https://huggingface.co/>, Accessed: 2025-07-29, 2025.
- [In24] Informatik, G. f.: Green Coding - Gesellschaft für Informatik e.V, 2024, <https://gi.de/aktuelles/projekte/green-coding>, accessed: 08/16/2024.
- [Ju23] Junger, D. et al.: Potentials of Green Coding - Findings and Recommendations for Industry, Education and Science. In: INFORMATIK 2023 - Designing Futures: Zukünfte gestalten. Gesellschaft für Informatik e.V., Bonn, pp. 1289–1299, 2023.
- [Ju25a] Junger, D. et al.: Academic Guidelines and Recommendations for Integrating Green Coding Aspects into Computer Science Curricula of Higher Education. In: Recent Advances in Environmental Science from the Euro-Mediterranean and Surrounding Regions (6th Edition). Accepted for publication, Marrakesh, Morocco, 2025.

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<sup>2</sup> <https://www.isocfoundation.org/> [2025-06-06]

- [Ju25b] Junger, D. M.: Mind Map Requirements Study for a Professional Green Coding Course Format, 2025, <https://doi.org/10.5281/zenodo.15728863>.
- [Ju25c] Junger, D. M.: Requirements Study for a Professional Green Coding Course Format - Supplementary Material A, 2025, <https://doi.org/10.5281/zenodo.15729451>.
- [JW24] Junger, D.; Wohlgemuth, V.: Bridging the Skills Gap: Insights from Professional Software Developers into the Challenges and Opportunities Faced by Young Career Starters Integrating Green Coding into Practice, Bonn, 2024.
- [Ki95] Kitzinger, J.: Qualitative Research: Introducing focus groups. *BMJ* 311 (7000), pp. 299–302, 1995, eprint: <https://www.bmj.com/content>, <https://www.bmj.com/content/311/7000/299>.
- [Mi25] Microsoft: Copilot (GPT-4) [Large Language Model], <https://copilot.microsoft.com/>, Accessed: 2025-07-29, 2025.
- [MM] MMG Connect: Focus Group Brief, <http://www.mmgconnect.com/projects/userfiles/file/focusgroupbrief.pdf>, Accessed: 2025-07-29.
- [NKD14] Naumann, S.; Kern, E.; Dick, M.: Classifying Green Software Engineering - The GREENSOFT Model. *Softwaretechnik-Trends* 33, pp. 18–19, 2014.
- [Ra14] Developing a sustainability non-functional requirements framework, GREENS 2014, Hyderabad, India: Association for Computing Machinery, 2014, pp. 1–8, <https://doi.org/10.1145/2593743.2593744>.
- [Ro20] Roberts, R.: Qualitative Interview Questions: Guidance for Novice Researchers. The Qualitative Report, 2020.
- [St24] Stack Overflow and Amazon: Levels of Formal Education for Software Developers, Worldwide, as of 2024 [Graph], 2024, <https://www.statista.com/statistics/793568/worldwide-developer-survey-level-formal-education/>, accessed: 08/20/2024.
- [Tu25] TurboScribe: TurboScribe - Unlimited audio & video transcription, 2025, <https://turboscribe.ai/>, accessed: 06/05/2025.